

Chronological Benchmarking of Tabular Regression Models for Paris Apartment Valuation Using DVF Transactions, 2021–2025

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August 2026

Abstract

This study benchmarks six methods for automated valuation of ordinary Paris apartments using geolocated French Demandes de valeurs foncières (DVF) transactions from 2021 through 2025. After mutation-level filtering and fixed quality controls, the dataset contains 143,009 sales. Model selection uses 90,275 observations from 2021–2023 for development training and 24,404 sales from 2024 for chronological validation. Selected configurations are refitted on 114,679 observations from 2021–2024 and evaluated on a common 2025 cohort of 28,330 transactions. The comparison includes an arrondissement median-price-per-square-metre heuristic, ridge regression, random forest, histogram gradient boosting, XGBoost, and LightGBM. LightGBM obtains the lowest 2025 mean absolute error (MAE), EUR 96,981, with $R^2 = 0.8619$ and a median absolute percentage error of 12.33%. XGBoost is statistically indistinguishable at the 5% level under a paired bootstrap: the LightGBM-minus-XGBoost MAE difference is EUR -387 with a 95% interval of approximately EUR $[-755, 1]$. All four nonlinear tree ensembles substantially outperform the market heuristic, whereas ridge regression performs poorly. The results support gradient-boosted trees as the practical reference class for this feature set, but do not establish that LightGBM universally dominates XGBoost or future market regimes.

Keywords: automated valuation model, real estate, DVF, temporal validation, gradient boosting, LightGBM, XGBoost, Paris.

1 Introduction

Automated valuation models (AVMs) estimate property transaction values from structured characteristics and market evidence. They are attractive because they are reproducible and scalable, yet they are vulnerable to spatial heterogeneity, missing property attributes, heavy-tailed prices, and temporal market shifts. A random train–test split can conceal the last problem because near-contemporaneous transactions appear on both sides of the split.

This work asks a focused question: under a shared chronological protocol, which of six common tabular-regression approaches provides the strongest out-of-year valuation accuracy for Paris apartments? The contribution is empirical rather than algorithmic. It consists of (i) mutation-aware DVF cleaning, (ii) a fixed 2021–2025 temporal evaluation design, (iii) model selection using a common criterion, (iv) paired uncertainty estimates on identical 2025 transactions, and (v) reproducible accuracy and computational-efficiency artifacts.

2 Data

2.1 Source and scope

DVF is produced by the French Direction générale des Finances publiques and contains transaction information derived from notarial deeds and cadastral records [3]. This study uses the geolocated

Paris files for calendar years 2021–2025. The raw files contain 420,058 property rows representing 207,365 mutations.

DVF records a mutation-level value on one or more property rows. Treating every row as an independent sale can therefore duplicate targets or associate a single price with incompatible property types. The pipeline retains arm’s-length sales (“Vente”) containing exactly one apartment (DVF local-type code 2), permits bundled dependencies such as a cellar or parking space (code 3), and rejects mutations containing a house or commercial property (codes 1 or 4).

The remaining observations must contain a valid date, transaction value, surface, Paris postcode, latitude, and longitude. Fixed plausibility bounds are 9–300 m² for surface, EUR 50,000–10,000,000 for value, and EUR 2,000–30,000 per m². Coordinates are restricted to the Paris bounding box used by the application. Table 1 reports the resulting cohorts.

Table 1: Annual DVF row counts and eligible apartment transactions.

Year	Raw rows	Single-apartment sale rows	Clean model rows
2021	81,517	31,835	30,397
2022	98,308	34,566	33,080
2023	80,993	28,174	26,798
2024	74,800	25,595	24,404
2025	84,440	29,503	28,330
Total	420,058	149,673	143,009

2.2 Features and target

The target is total mutation value in euros. Learned models optimize $\log(1 + y)$ and transform predictions back with $\exp(\hat{z}) - 1$, reducing the influence of the long upper tail while preserving a positive-scale prediction. The feature vector contains built surface, number of principal rooms, latitude, longitude, address number, number of lots, month, day of year, calendar year, months since January 2021, and postcode. Numeric missing values are median imputed; postcode is one-hot encoded. Numeric inputs are standardized only for ridge regression, where scale directly affects the penalty.

Important determinants are unavailable in DVF: interior condition, floor, elevator, energy rating, exposure, view, occupancy, and detailed building quality. The estimated error therefore reflects both model limitations and irreducible missing-information error.

3 Methods

3.1 Temporal evaluation protocol

The split is chronological and identical for every method:

- development training: 2021–2023 (90,275 rows);
- configuration selection: 2024 (24,404 rows), minimizing MAE;
- final refit: 2021–2024 (114,679 rows);
- common test: 2025 (28,330 rows).

No 2025 transaction is used by a fitting or configuration-selection call in the benchmark program. However, 2025 results from an earlier baseline experiment had already been inspected by the analyst.

Consequently, 2025 is computationally held out for the six-model comparison but should not be described as a fully blind external test.

3.2 Models

The median-price baseline multiplies apartment surface by the median price per m² observed in its arrondissement during the available training years. Ridge regression adds an ℓ_2 penalty to a linear model [5]. Random forest averages decorrelated trees trained on bootstrap samples and randomized feature subsets [1]. Histogram Gradient Boosting is the existing project model and represents conventional histogram-binned boosting [4]. XGBoost adds a regularized, systems-optimized tree-boosting implementation [2]; LightGBM uses histogram learning and leaf-wise growth with efficiency techniques including gradient-based one-side sampling and exclusive feature bundling [6].

Table 2 gives the selected configuration. Each learned model is chosen from a compact, predefined search space of three candidates, except ridge regression, which uses four penalty values. This is a controlled engineering benchmark, not an exhaustive hyperparameter optimization.

Table 2: Configurations selected by 2024 validation MAE. Shared random seed is 42.

Method	Selected configuration
Median price/m ²	Arrondissement median from the training period
Ridge	$\alpha = 100$
Random Forest	240 trees; max-features 0.7; minimum leaf size 5
Histogram Gradient Boosting	63 leaves; minimum leaf size 40; $\ell_2 = 2$; learning rate 0.06
XGBoost	depth 6; minimum child weight 5; 650 trees; learning rate 0.04
LightGBM	31 leaves; minimum child size 25; 650 trees; learning rate 0.04

3.3 Evaluation criteria

For actual values y_i and predictions $\hat{y}_i$, the primary criterion is

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|. \quad (1)$$

Secondary criteria are root mean squared error (RMSE), coefficient of determination (R^2), mean absolute percentage error (MAPE), median absolute percentage error (MdAPE), and the proportions within 10% and 20% of the actual value. Training time, batch prediction latency, and serialized artifact size are also recorded.

Individual MAE intervals use 500 nonparametric bootstrap resamples. Model-to-model conclusions use 1,000 paired bootstrap resamples of per-transaction absolute-error differences, preserving the fact that all models predict the same sales. A paired interval wholly below zero indicates that the row model has lower MAE at the 5% level under this resampling procedure.

4 Results

4.1 Overall accuracy

Table 3 presents the common 2025 test results. LightGBM has the lowest observed MAE at EUR 96,981, followed by XGBoost at EUR 97,367 and Histogram Gradient Boosting at EUR 97,753. The same three models are almost tied on typical relative error, with MdAPE between 12.33% and 12.48%. LightGBM puts 71.41% of test predictions within 20% of the transaction value, compared with 62.60% for the market baseline.

Table 3: Performance on 28,330 transactions from the common 2025 test cohort.

Model	MAE (EUR)	95% MAE interval	RMSE (EUR)	R^2	MdAPE	Within 10%	Within 20%
LightGBM	96,981	[94,935; 99,014]	197,107	0.8619	12.33%	41.49%	71.41%
XGBoost	97,367	[95,312; 99,485]	199,023	0.8592	12.34%	41.22%	71.32%
Hist. Gradient Boosting	97,753	[95,730; 99,884]	198,027	0.8606	12.48%	40.87%	70.87%
Random Forest	99,797	[97,726; 101,900]	200,909	0.8566	12.89%	40.20%	70.06%
Median price/m ²	110,868	[108,575; 113,272]	221,015	0.8264	14.62%	36.54%	62.60%
Ridge regression	169,126	[163,427; 174,924]	549,280	-0.0722	19.99%	25.51%	50.03%

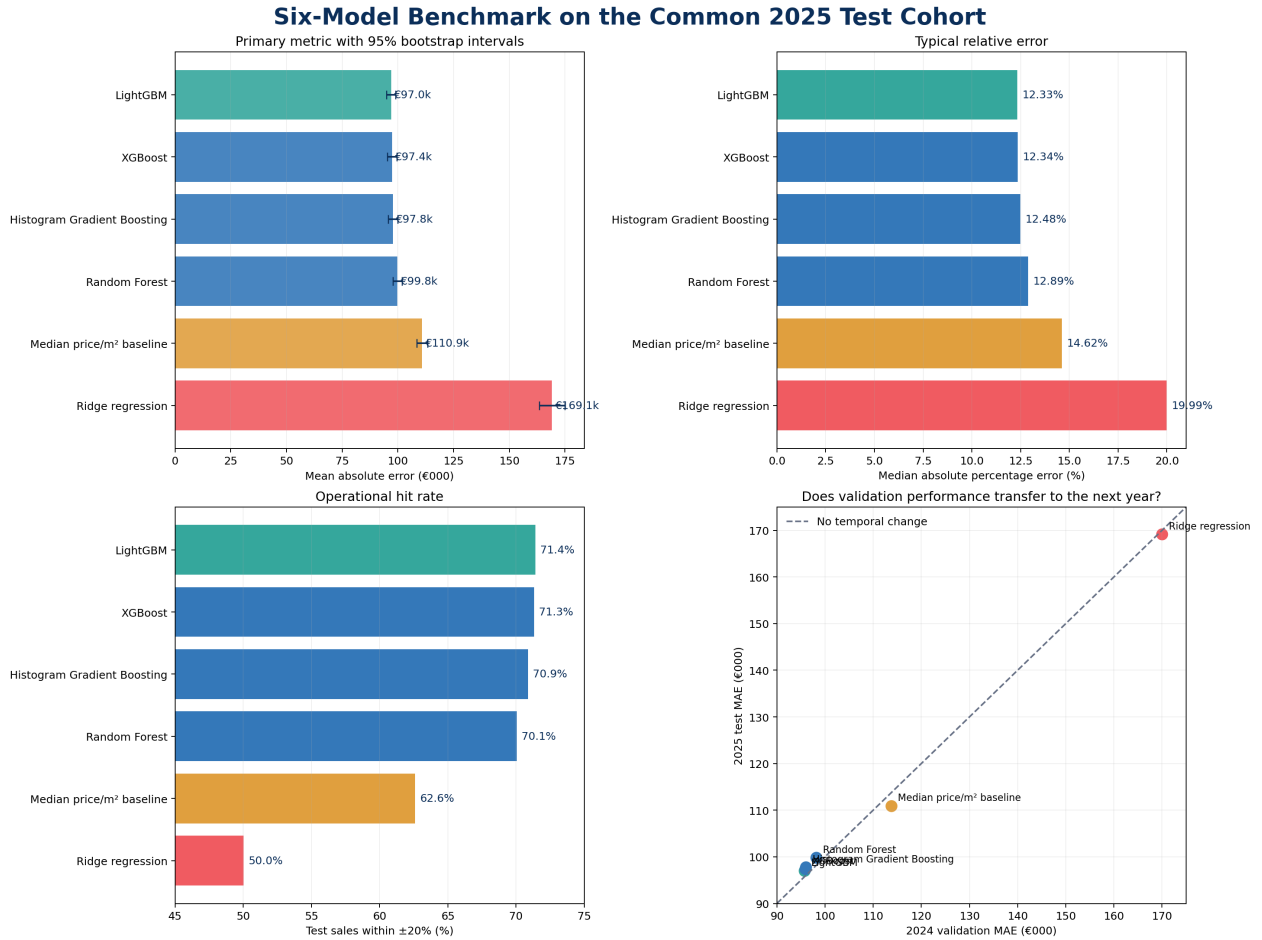


Figure 1: Accuracy comparison, operational hit rate, and validation-to-test transfer. Error bars in the MAE panel are 95% bootstrap intervals.

Relative to the median-price baseline, LightGBM reduces test MAE by EUR 13,887 (paired 95% interval EUR 12,896–14,905), a 12.53% reduction. The corresponding reductions are EUR 13,501 for XGBoost, EUR 13,115 for Histogram Gradient Boosting, and EUR 11,071 for Random Forest; their paired intervals against the baseline are all strictly positive.

4.2 Are the leading models distinguishable?

The LightGBM-minus-XGBoost paired MAE difference is EUR -387 with a 95% interval of EUR $[-755, 1]$. Thus, the observed ordering does not support a 5%-level claim that LightGBM is more accurate than XGBoost. LightGBM is more accurate than Histogram Gradient Boosting by EUR 772 per sale, with interval EUR $[400, 1,140]$ in LightGBM’s favor, and more accurate than Random Forest by EUR 2,816, with interval EUR $[2,257, 3,331]$. Figure 2 reports the full paired difference matrix.

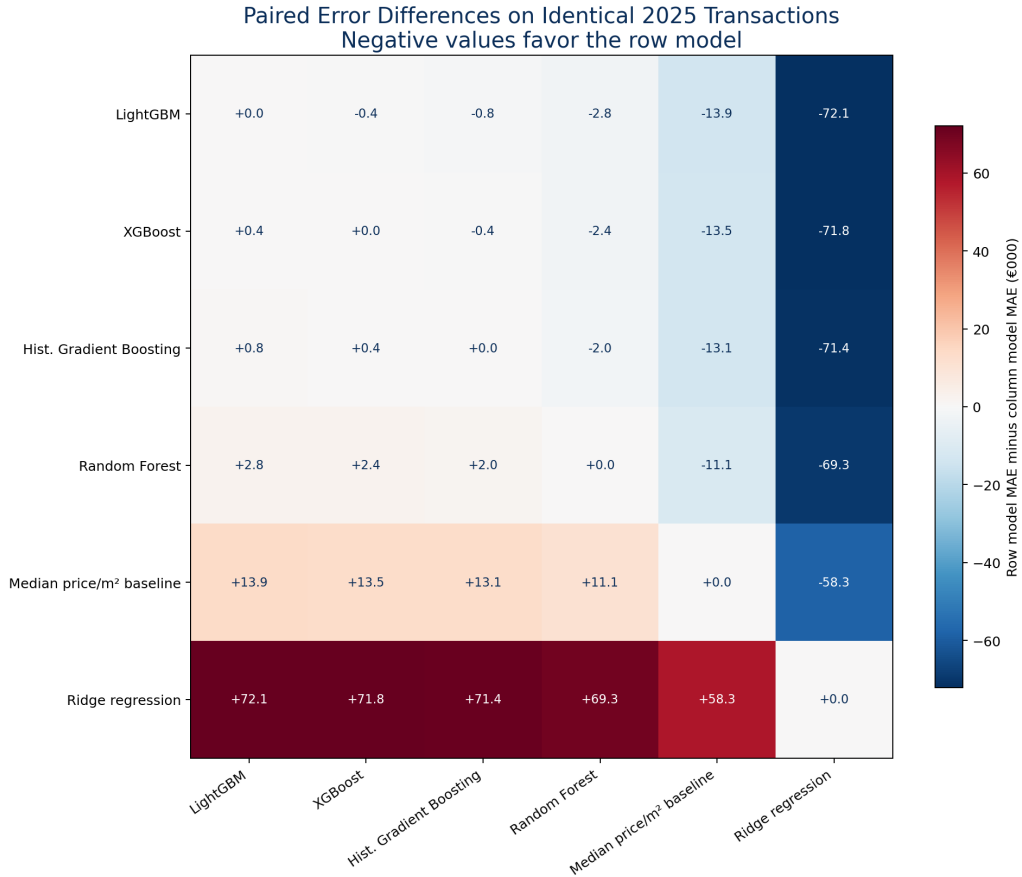


Figure 2: Mean paired absolute-error differences in thousands of euros. Negative values favor the row model.

4.3 Efficiency and temporal stability

On the benchmark machine, the selected LightGBM refit took 2.05 seconds and its serialized artifact occupied 1.84 MB. XGBoost took 3.27 seconds and occupied 2.60 MB; Histogram Gradient Boosting took 2.87 seconds and occupied 1.36 MB. Random Forest required 16.63 seconds and 383.27 MB. These single-run wall-clock measurements describe this implementation and hardware, not universal library speed rankings.

All nonlinear ensembles retain a clear advantage over the baseline throughout 2025. The boosted models have monthly MdAPE near 12% in the first half and generally rise toward 13–14% by

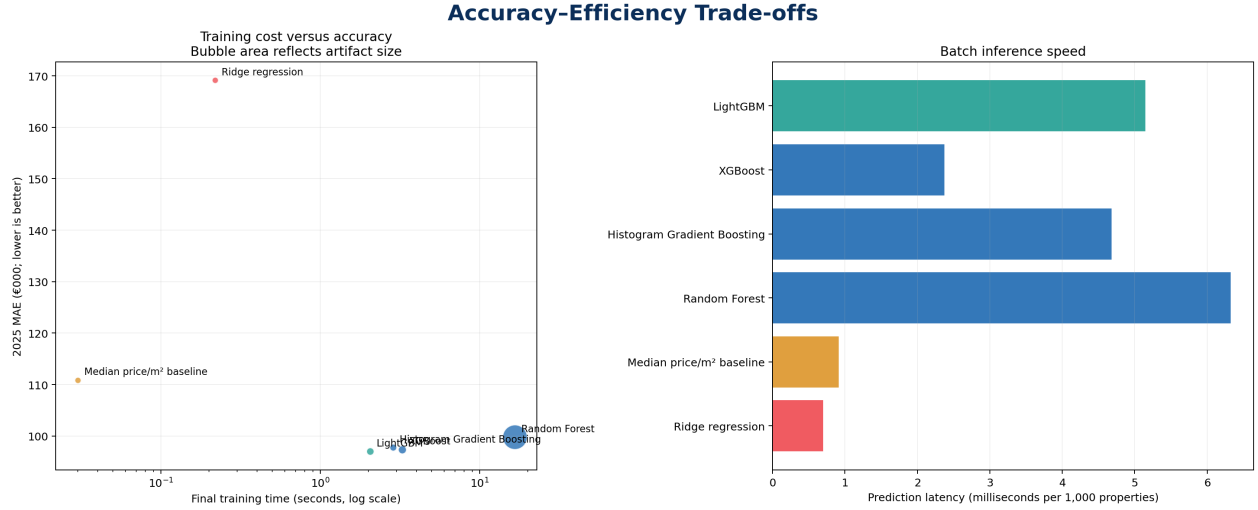


Figure 3: Observed accuracy–efficiency trade-offs. Training-time axes are logarithmic; bubble area represents serialized artifact size.

December, suggesting modest late-year drift. Ridge is consistently weakest and deteriorates more strongly in the second half.

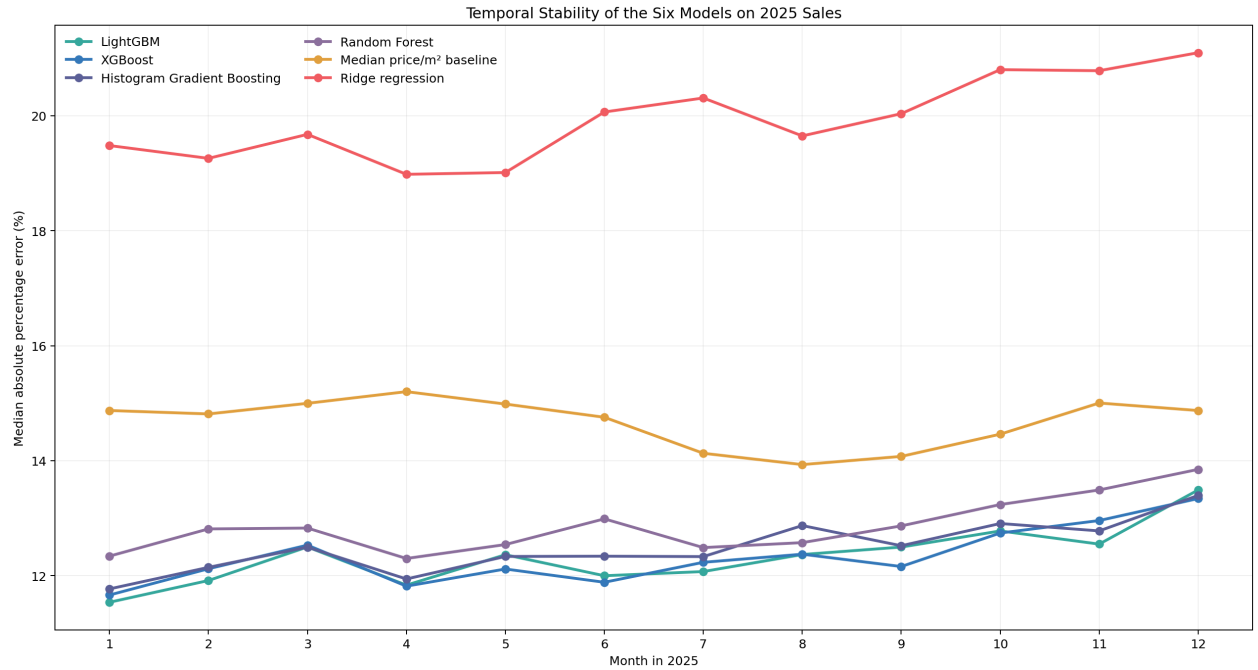


Figure 4: Month-by-month median absolute percentage error on the 2025 test cohort.

5 Discussion

The most important result is class-level rather than library-level: nonlinear tree ensembles are clearly preferable to the market heuristic and the global linear model for the available DVF features. Location coordinates, postcode, surface, and their interactions create irregular valuation functions that ridge regression cannot represent without explicit basis expansion. Its negative test R^2 and high RMSE also indicate sensitivity to expensive-tail transactions after retransformation from log space. Among tree ensembles, gradient boosting is the strongest family. LightGBM is the best point estimate

and offers an attractive accuracy–size–training-time combination. Nevertheless, its difference from XGBoost is too small to declare a robust accuracy victory on this test cohort. A production choice between the two should therefore consider operational simplicity, explainability tooling, latency under the actual service workload, and monitoring behavior in newer market periods.

The strong arrondissement median-price baseline is also informative. It reaches $R^2 = 0.8264$ using only surface and arrondissement-level market evidence. The boosted models improve MAE by approximately 12%–13%, demonstrating genuine incremental value, but the remaining EUR 97k MAE shows that the current feature set is not sufficient for appraisal-grade precision.

6 Limitations and validity threats

First, the 2025 cohort is temporally held out by the benchmark code but was previously inspected during development of the original project model. It is not an untouched external replication. A future publication claim should be tested on a newly released period or another geography that has not informed model or feature decisions.

Second, DVF does not contain many causal value drivers, and mutation-level prices may include dependencies. The eligibility rules reduce ambiguity but do not recover unobserved condition or amenity information. Third, the search is compact and slightly different in size across model families. It is designed to avoid a large, unfair tuning advantage, but it does not locate each method’s global optimum. Fourth, bootstrap intervals condition on the observed 2025 transaction population and do not quantify regime-change uncertainty. Finally, timing values come from one Windows workstation and one run per selected final model.

7 Reproducibility

The experiment uses Python 3.12, NumPy 1.26.4, pandas 2.2.2, scikit-learn 1.5.1, XGBoost 3.4.0, LightGBM 4.7.0, and joblib 1.4.2. Random seeds are fixed at 42 for model fitting and 20250811 for bootstrap resampling. The complete benchmark took 294.1 seconds in the recorded run. The project stores the cleaning and benchmark code, annual input manifest, selected model artifacts, per-row validation and test predictions, hyperparameter-search results, pairwise comparisons, and all figures required by this paper.

The principal commands are:

```
python benchmark_models.py
python visualize_benchmark.py
powershell -ExecutionPolicy Bypass -File docs/paper/compile_paper.ps1
```

8 Conclusion

Under a shared chronological evaluation, LightGBM produces the lowest observed 2025 MAE, while XGBoost is statistically tied under paired resampling. Histogram Gradient Boosting and Random Forest remain competitive, and all four nonlinear ensembles outperform the arrondissement median-price baseline. Ridge regression is unsuitable as the principal AVM with the current representation. The next scientific step is not to assume that a more complex architecture will win, but to add richer building and neighborhood information and benchmark any attention- based model against the LightGBM/XGBoost reference under a newly locked temporal or geographic test.

Acknowledgements

The computational workflow and manuscript preparation were assisted by an AI coding system. Responsibility for data use, analysis choices, verification, and all scientific claims remains with the project author.

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